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Estimability of recharge through groundwater model calibration: Insights from a field-scale steady-state example

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ABSTRACT

The ability of groundwater models to inform recharge through calibration is hampered by the correlation between recharge and aquifer parameters such as hydraulic conductivity (K), and the insufficient information content of observation datasets. These factors collectively result in non-uniqueness of parameter estimates. Previous studies that jointly estimate spatially distributed recharge and hydraulic parameters are limited to synthetic test cases and/or do not evaluate the effect of non-uniqueness. The extent to which recharge can be informed by calibration is largely unknown for practical situations, in which complexities such as parameter heterogeneities are inherent. In this study, a systematic investigation of recharge, inferred through model calibration, is undertaken using a series of numerical experiments that include varying degrees of hydraulic parameter information. The analysis involves the use of a synthetic reality, based on a regional-scale, highly parameterised, steady-state groundwater model of Uley South Basin, South Australia. Parameter identifiability is assessed to evaluate the ability of parameters to be estimated uniquely. Results show that a reasonable inference of recharge (average recharge error <10%) requires a surprisingly large number of preferred value regularisation constraints (>100 K values across the 129 km^2 study area). The introduction of pumping data into the calibration reduces error in both the average recharge and its spatial variability, whereas submarine groundwater discharge (as a calibration target) reduces average recharge error only. Nonetheless, the estimation of steady-state recharge through inverse modelling may be impractical for real-world settings, limited by the need for unrealistic amounts of hydraulic parameter and groundwater level data. This study provides a useful benchmark for evaluating the extent to which field-scale groundwater models can be used to inform recharge subject to practical data-availability limitations.

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1. Introduction

Reliable estimates of recharge and its variability in time and space are often a requirement for effective groundwater management. This is particularly the case where recharge estimates are used as a basis for allocating groundwater extraction rates (Werner et al., 2011). Traditionally, recharge is inferred from groundwater-level changes (e.g., Crosbie et al., 2005), solute concentrations (e.g., Wood and Sanford, 1995), unsaturated zone numerical modelling (e.g., Keese et al., 2005) and/or remote sensing techniques (e.g., Jackson, 2002). The reliability of these approaches depends on many factors, including site characteristics (e.g., climate, vegetation, soil, topography, depth to watertable) and the availability and accuracy of field data (Scanlon et al., 2002). Estimates derived from field measurements also require consideration of the representative spatial and temporal scales (Scanlon et al., 2002). For these reasons, recharge is widely regarded as one of the most challenging water-balance components to quantify (Dripps and Bradbury, 2007).

An alternative method of recharge estimation is inverse modelling using numerical groundwater models. This involves the inference of recharge through calibration or "history matching" (i.e., minimising the discrepancy between field observations and corresponding model-generated outputs). The application of groundwater models in this context is appealing because of their ability to account for important nonlinear interactions between recharge, discharge, evapotranspiration and changes in groundwater storage (Sanford, 2002). Additionally, recharge values estimated

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from groundwater models typically apply to regional scales (e.g., >25 km^2), which are directly applicable to management decision making. Stoertz and Bradbury (1989) presented one of the first attempts at informing recharge by groundwater model calibration. They produced a map of recharge and discharge zones, and provided a range of spatially averaged net recharge rates for a basin in Wisconsin (USA). Groundwater model calibration has since been used frequently to inform recharge (e.g., D'Agnese et al., 1999; Essaid et al., 2003; Lubczynski and Gurwin, 2005; Palma and Bentley, 2007; Yidana, 2011; Hashemi et al., 2013). However, the uncertainty associated with groundwater model-based recharge estimates is rarely quantified.

The primary challenge associated with recharge estimation through groundwater model calibration lies in the correlation between recharge and hydraulic properties such as hydraulic conductivity (K) and specific yield (Scanlon et al., 2002). Parameter correlation leads to non-uniqueness of estimated parameter values (Carrera and Neuman, 1986; McKenna et al., 2003). That is, multiple parameters can be varied in such a way that a model's outputs (such as the heads used in a calibration objective function) are unaffected. Non-uniqueness arises from the limited information content of observation datasets used as a basis for calibration. In the case where only groundwater-level observations are available, a unique solution for recharge and K cannot be obtained; only their ratio can be estimated uniquely (Scanlon and Cook, 2002). Efforts to overcome parameter non-uniqueness for the purposes of estimating recharge through calibration have involved several approaches (Sanford, 2002), including the use of flux observations (e.g., stream base flow; Arnold et al., 2000; Hunt et al., 2006) or seepage velocities (e.g., based on groundwater age; Portniaguine and Solomon, 1998; Sanford et al., 2004), either as prescribed stresses or as calibration targets.

An alternative means to address groundwater model non-uniqueness is by imparting expert knowledge of aquifer hydraulic properties, such as K, through regularisation (Carrera et al., 2005). Regularisation refers to the stabilisation of the parameter estimation process such that a unique solution to an otherwise ill-posed inverse problem can be obtained (Engl et al., 1996). For example, Tikhonov regularisation schemes involve the specification of preferred parameter values and/or relationships based on expert knowledge to supplement the information contained within the calibration dataset. Then, any deviation in parameters from preferred values will occur only to the extent that is necessary to achieve a target level of model-to-measurement misfit (Tikhonov and Arsenin, 1977; Doherty and Skahill, 2006). Theoretically, a unique estimate of recharge may be achieved providing adequate constraints are imparted on K parameter values and relationships; however, the extent to which this is the case, for practical model calibration applications, is not presently understood. Aquifer hydraulic parameter distributions are often assumed to be homogeneous or piecewise homogeneous for the purposes of informing recharge (e.g., Sanford et al., 2004; Gómez et al., 2010; Cao et al., 2013). These approaches assume that any errors in recharge arising from the simplification of hydraulic parameter distributions are small (e.g., Ireson and Butler, 2013).

The uncertainty in various model input parameters, in particular recharge and K, has led to "simultaneous" or "joint" estimation of such parameters. For example, Hendricks Franssen et al. (2004) jointly estimated spatially variable transmissivity and recharge for a simple synthetic aquifer in their study of the impact of transmissivity and recharge uncertainty on well capture zones. They highlighted the importance of regularisation for obtaining reasonable parameter estimates. This was also demonstrated by Friedel (2005) in an investigation of the effect of variations in calibration dataset information on parameter non-uniqueness for a synthetic vadose zone model. Despite these efforts, Hendricks Franssen et al. (2009) subsequently asserted that very few studies estimate multiple parameter types simultaneously within complex, regional-scale groundwater models, and that this constituted an important direction for future research.

A number of studies that jointly estimate recharge and aquifer hydraulic parameters for real-world settings have been presented. For example, Hendricks Franssen et al. (2008) used a stochastic calibration procedure to estimate spatial distributions of recharge and K in a groundwater model of the Chobe region, Botswana (Africa). They evaluated the value of remote sensing data and other data combinations by considering changes to ensemble recharge and K distribution statistics. However, their study did not investigate the influence of K constraints on calibrated estimates of recharge, given that stochastic methods are less likely to display parameter surrogacy (i.e., the process whereby parameters take on spurious values to compensate for other model errors; Clark and Vrugt, 2006; White et al., 2014). Hayley et al. (2014) estimated spatial recharge and K distributions simultaneously in their calibration of a highly parameterised groundwater model of north-eastern Alberta (Canada) using cloud computing. However, their calibrated recharge and K fields were evaluated separately in terms of reasonableness. That is, the effect of parameter non-uniqueness was not considered in their discussion of calibration outcomes. Carniato et al. (2015) jointly estimated the spatial distribution of recharge, K and specific yield as part of their development of a reactive transport model of a groundwater contamination field site in northern Belgium. They demonstrated that accurate predictions of plume concentrations and reduced uncertainty associated with recharge and K were obtained by incorporating both groundwater level and concentration data into the calibration. However, the degree to which their inverse modelling outcomes were influenced by parameter non-uniqueness was not evaluated.

Recently, Erdal and Cirpka (2016) applied the Ensemble Kalman filter (Evensen, 1994) to jointly estimate spatially distributed recharge and K on the basis of head observations for a highly simplified synthetic aquifer. They showed that accurate values of recharge and K can be estimated simultaneously providing that a sufficient degree of prior information is available (i.e., the geostatistics of the true (initial ensemble) recharge- and K-fields are well-known). However, they discussed that their findings regarding the amount of prior information required for accurate recharge estimation may not be transferable to real-world applications, and recommended that more realistic cases be investigated in future studies to provide insights into the estimability of recharge and hydraulic parameters for practical situations.

Presently, the extent to which groundwater models can be used to estimate recharge through calibration, for complex, field-scale aquifer settings, is largely unknown. This is despite the generally accepted notion that groundwater models can be used to estimate recharge provided that groundwater level distributions and other model parameters such as K are known "well enough", as advocated by Sanford (2002).

The objective of this study is to evaluate the ability of groundwater model calibration to inform recharge in the face of non-uniqueness by using a systematic inverse modelling approach involving a regional-scale, highly parameterised groundwater model of Uley South Basin (USB), South Australia. We intentionally use the most common approach to groundwater model calibration, namely the application of gradient methods such as those adopted in PEST (Doherty, 2016) and UCODE (Poeter and Hill, 1998), so that the findings offer guidance to groundwater practitioners. A quasi-hypothetical modelling strategy, whereby a reference model that serves as a synthetic reality for comparing subsequent model results, is adopted. The current assessment focuses on the spatial distribution of time-averaged recharge and basin-wide recharge volumes. This is achieved based on steady-state modelling, as is

considered in many previous investigations of recharge (e.g., Gómez-Hernández and Gorelick, 1989; Zhang, 2014). The USB is chosen as a case study because of the well-developed understanding of the regional flow system based on extensive observation datasets, and the availability of an existing highly parameterised groundwater model developed by Knowling et al. (2015). Also, spatially distributed recharge rates are available from field-based estimates and one-dimensional unsaturated zone modelling (Ordens et al., 2012; Ordens, 2014).

2. Theoretical background

Two-dimensional Dupuit-Forcheimer flow within a heterogeneous aquifer under steady-state conditions is described by the diffusion equation:

$$\frac{\partial}{\partial x}\left(K_x b \frac{\partial h}{\partial x}\right) + \frac{\partial}{\partial y}\left(K_y b \frac{\partial h}{\partial y}\right) + W = 0 \quad (1)$$

where x and y [L] are Cartesian coordinates, h [L] is the hydraulic head, b [L] is the saturated aquifer thickness (which varies with h in unconfined aquifer systems), W [L/T] is the volumetric source/sink term per unit area that expresses the combined effects of groundwater recharge, pumping, leakage, etc. (positive W values indicate inflow), and K_x and K_y [L/T] are the hydraulic conductivities in the x and y directions, respectively.

Eq. (1) is solved in a piecewise manner by groundwater models that use numerical methods such as finite-difference or finite-element approximations (e.g., Harbaugh et al., 2000; Diersch, 2005). At the node scale, the computation of h is a function of the ratio of recharge-to-K (i.e., W/K, considering recharge as the only sink/source component for simplicity). It follows that a unique estimate of recharge and K may not be attainable based on h observations alone, and that only the ratio W/K may be determinable without additional information pertaining to K or groundwater fluxes (e.g., $Kb\frac{\partial h}{\partial x}$). Non-uniqueness at the scale of the model domain is complicated by such aspects as the spatial patterns in K, flow directions, boundary conditions, etc., and is therefore more challenging to characterise for practical situations, which are hampered by observation-data scarcity. The current study focuses on the estimability of recharge in the face of non-uniqueness for practical situations, in response to the knowledge gap identified by Hendricks Franssen et al. (2009).

3. Methodology

3.1. Study area: Uley South Basin (South Australia)

The USB (34°76′ S, 135°56′ E) is a topographically enclosed coastal groundwater basin of area ~129 km^2, and is located in the southern Eyre Peninsula, South Australia. The region has a semi-arid climate, which is characterised by winter-dominant rainfall (May–October), and hot, dry summer months (November–April) (Harrington et al., 2006). Long-term average annual rainfall and pan evaporation rates are 560 and 1547 mm/y, respectively (Bureau of Meteorology, 2010). The land surface is composed mainly of skeletal soils of sandy and clayey loam (Evans, 1997) and exposed calcrete, where a widespread distribution of sinkholes serve to redistribute surface water into the unsaturated zone (Ordens et al., 2012). USB's vegetation consists primarily of various scrub species, Mallee trees, drooping she-oak and significant areas of sparse grassland (Bresciani et al., 2014). Groundwater in USB occurs predominantly within an unconfined Quaternary sand and limestone aquifer (QL), which is underlain by a discontinuous Tertiary clay aquitard (TC) and a semi-confined Tertiary aquifer comprising silty and clayey sand (TS) (Zulfic et al., 2007). Groundwater abstraction occurs from the QL aquifer, which constitutes the primary freshwater supply for the Eyre Peninsula region (Harrington et al., 2006). Historically, allowable extraction rates have been based on a fixed proportion of volumetric recharge estimates (EPNRM, 2006). Despite the karst nature of some of the USB sediments, the aquifer behaves mostly as a single-porosity (i.e., equivalent porous medium) setting, at least at scales of interest to water resources management and regional-scale modelling (Werner, 2014).

3.2. Groundwater model

The groundwater model of USB used as a basis for the current study is described in detail by Knowling et al. (2015). MODFLOW (Harbaugh et al., 2000) is used to simulate groundwater flow within USB under steady-state conditions. The assumption of steady-state conditions is considered a reasonable approximation of the average groundwater levels during two distinct time intervals: 1960–1976 (i.e., pre-development) and 2005–2009 (i.e., a period of relatively stable water levels). The pre-development steady-state condition is adopted in the reference USB model. The contemporary condition (2005–2009), which involves significant groundwater extraction, serves as a basis for developing an alternative steady-state model. This model is used to explore whether or not the addition of prescribed fluxes (i.e., groundwater extraction) assists in constraining the estimation of recharge. Also, considering two alternative steady-state models allows for an evaluation of whether or not the estimability of recharge is specific to a particular steady-state condition. Table 1 lists the main features of the reference USB model. Fig. 1 shows a schematic of the reference model and its parameter distributions.

The spatial variability of K is achieved using 125 pilot points (de Marsily et al., 1984) distributed in a regular grid configuration (Fig. 1a). Cell-by-cell K variability is specified by kriging based on an isotropic, exponential variogram of log(K), with a sill of one, a range of 1200 m, and a nugget of zero. A single boundary conductance (C) is assigned to each GHB reach (Fig. 1a). It should be noted that the reference model K and C distributions (Fig. 1b–d) differ slightly from those obtained by Knowling et al. (2015). They estimated K and C parameters through joint calibration of the pre-development and contemporary steady-state models, whereas reference model K and C distributions are estimated on the basis of only the pre-development model in the current study.

3.3. Model calibration

Estimation of groundwater model parameters is undertaken using PEST (Doherty, 2016). BeoPEST (Schreüder, 2009), a version of PEST that allows model-run parallelisation, is used to reduce computational times. PEST adopts a gradient-based algorithm (i.e., the Gauss-Marquardt-Levenberg (GML) method; Levenberg, 1944) to minimise a weighted least-squares objective function, Φ. For parameter estimation problems involving Tikhonov regularisation, Φ is given by:

$$\Phi = (\mathbf{Xp} - \mathbf{h})^t \mathbf{Q}_m (\mathbf{Xp} - \mathbf{h}) + \mu(\mathbf{Zp} - \mathbf{d})^t \mathbf{Q}_r (\mathbf{Zp} - \mathbf{d}) \quad (2)$$

where $\mathbf{p}$ is a vector containing model parameter values, $\mathbf{h}$ is a vector containing measurement observation values, $\mathbf{X}$ is a sensitivity (or "Jacobian") matrix containing derivatives of model outputs (for which there are corresponding measurement observations $\mathbf{h}$) with respect to $\mathbf{p}$, $\mathbf{d}$ is a vector containing Tikhonov regularisation "observations" (representing preferred parameter values and/or relationships), $\mathbf{Z}$ is a Jacobian matrix containing derivatives of model-generated counterparts to regularisation observations $\mathbf{d}$ with respect to $\mathbf{p}$, μ is a regularisation weight factor (discussed below), and

Table 1
Summary of reference model characteristics.

Model characteristic	Description
Layer(s) type	Unconfined upper layer (QL aquifer); Convertible confined/unconfined (i.e., depending on hydraulic head relative to layer top elevation) lower layer (TS aquifer); TC aquitard simulated implicitly, i.e., a "quasi-3D approach" (Chiang and Kinzelbach, 1998)
Spatial discretisation	100 m × 100 m model-grid cells (245 rows, 132 columns; Fig. 1a)
K	Estimated through calibration (Fig. 1b and c) and constrained by field data (e.g., pumping test results) or literature values. Vertical K assumed to be 0.1 times horizontal K
Ocean boundary	Constant-head boundary using density corrections following Morgan et al. (2012)
Interaction with adjacent basins	General-head boundaries (GHB; Harbaugh et al., 2000) used. GHB forcing heads are based on time-averaged groundwater levels in adjacent basins. GHB conductances (C) estimated through calibration (Fig. 1d), and constrained by knowledge of K and aquifer thickness
Spatially distributed recharge	Applied to uppermost active layer. Recharge distributions based on the modelling by Ordens (2014) (e.g., Fig. 1e). Spatially averaged recharge for pre-development conditions is 100 mm/y, and for contemporary conditions is 69 mm/y
Evapotranspiration (ET)	ET package (EVT; Harbaugh et al., 2000) simulates shallow groundwater ET, whereas unsaturated zone ET is accounted for in the recharge modelling of Ordens (2014)
Groundwater pumping	Pumping rates at each well based on metered data and extrapolation (where metered data are unavailable) (Werner et al., 2011). No pumping occurs during pre-development conditions. Total pumping rate for contemporary conditions is 6.98×10^6 m^3/y

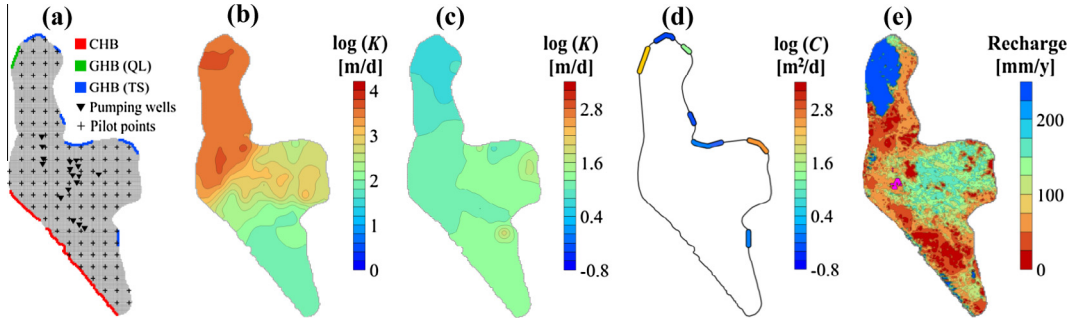


Fig. 1. Reference model characteristics: (a) model domain, grid, boundary conditions ("CHB" denotes constant-head boundary, and "GHB" denotes general-head boundary), extraction well and pilot point locations; (b) log(K) [m/d] of the QL aquifer; (c) log(K) [m/d] of the TS aquifer; (d) log(C) [m^2/d] for each GHB reach, and (e) spatially distributed recharge rates [mm/y] (temporally averaged over the pre-development period 1960–1976) from Ordens (2014).

$\mathbf{Q}_m$ and $\mathbf{Q}_r$ are matrices whose diagonal elements are the square of the weights assigned to corresponding measurement and regularisation observations, respectively. The value of μ controls the trade-off between fitting observation data and the incorporation of regularisation constraints, explained in more detail below.

Traditional calibration approaches, which involve the inference of a small number of parameters, minimise the measurement objective function, Φ_m, which is the term $(\mathbf{Xp}-\mathbf{h})^t\mathbf{Q}_m(\mathbf{Xp}-\mathbf{h})$ in Eq. (2). Where the inverse problem is "well-posed" or "over-determined", minimisation of Φ_m yields a unique parameter set. The regularisation objective function, Φ_r, given by the term $(\mathbf{Zp}-\mathbf{d})^t\mathbf{Q}_r(\mathbf{Zp}-\mathbf{d})$ in Eq. (2), encapsulates the deviation of parameter values from those representing a preferred condition. A common application of Tikhonov regularisation is to include the difference between neighbouring pilot-point parameter values in $\mathbf{d}$. By specifying the corresponding elements of $\mathbf{d}$ equal to zero, a preferred homogeneity constraint is enforced (Doherty, 2003). The desired level of model-to-measurement misfit (Φ_m) is user-specified as the "target measurement objective function" Φ_m^l, which is obtained by adjusting μ. Φ_r is thereby minimised under the constraint that:

$$\Phi_m \leqslant \Phi_m^l \tag{3}$$

This process constitutes a constrained minimisation problem (e.g., deGroote-Hedlin and Constable, 1990), whereby Φ_m^l provides a user-specified basis for controlling the degree to which the problem is regularised. This term is prescribed in accordance with the uncertainty associated with observation measurements. The reader is referred to Fienen et al. (2009) for a detailed discussion on Φ_m^l and other PEST-specific regularisation control variables.

For regularised (non-linear) model calibration problems, a parameter upgrade vector, $\Delta\mathbf{p}$, as calculated iteratively using the GML procedure, is given by (Doherty and Skahill, 2006):

$$\Delta\mathbf{p} = (\mathbf{X}^t\mathbf{Q}_m\mathbf{X} + \mu\mathbf{Z}^t\mathbf{Q}_r\mathbf{Z} + \lambda\mathbf{I})^{-1}(\mathbf{X}^t\mathbf{Q}_m\mathbf{r} + \mu\mathbf{Z}^t\mathbf{Q}_r\mathbf{s}) \tag{4}$$

where λ is the Marquardt lambda parameter, $\mathbf{I}$ is the identity matrix, $\mathbf{r}$ is a vector of the model-to-measurement discrepancies $(\mathbf{Xp}-\mathbf{h})$, and $\mathbf{s}$ is a vector of the regularisation discrepancies $(\mathbf{Zp}-\mathbf{d})$. The λ term is often increased such that $\mathbf{X}^t\mathbf{Q}_m\mathbf{X}$ becomes invertible to prevent instability in the calculation of the parameter upgrade. Despite the ability of high λ values to impart stability, robust parameter estimation requires some form of regularisation (Tonkin and Doherty, 2005). The reader is referred to Doherty and Hunt (2010) and references cited therein for a more detailed description of pertinent calibration methodologies.

In this study, the sensitivities of model outputs to parameter values within the $\mathbf{X}$ and $\mathbf{Z}$ matrices are calculated using central finite differences with a 1% perturbation of current parameter values following, e.g., James et al. (2009). The maximum parameter value upgrade factor between optimisation iterations is 20% of current values. The specification of Φ_m^l is dependent on initial modelling results. Weighting of different regularisation observation groups is achieved on the basis of total composite observation-group sensitivities, which is the sum of the (weighted) sensitivities of all observations within a particular group to all adjustable parameters (Hill, 1998, 2010). Individual regularisation observations (i.e., within a group) are weighted uniformly. The initial λ value and the factor by which λ is adjusted during parameter upgrade

testing is set to 10 and −3, respectively, as suggested by Doherty (2016).

3.4. Parameter identifiability

Parameters are evaluated in terms of their identifiability (i.e., their ability to be estimated uniquely through calibration). The set of parameter values obtained through model calibration $\underline{\mathbf{p}}$ is represented by (Doherty et al., 2010):

$$\underline{\mathbf{p}} = \mathbf{G}\mathbf{h} \tag{5}$$

where **G** is a matrix (sometimes referred to as the "parameter solution matrix") expressing the means by which the estimated parameter values are obtained from measurement observations **h**. The formulation of **G** is therefore dependent on the regularisation method employed within the calibration process. In this study, **G** is given by the following (Doherty, 2016):

$$\mathbf{G} = (\mathbf{X}^t\mathbf{Q}_m\mathbf{X} + \mu\mathbf{Z}^t\mathbf{Q}_r\mathbf{Z} + \lambda\mathbf{I})^{-1}\mathbf{X}^t\mathbf{Q}_m \tag{6}$$

Eq. (6) applies to the situation of calibrated parameter values, and assumes that **r** and **s** are sufficiently small that they can be ignored. The relationship between a set of measurement observations **h** and a set of true (i.e., real-world) system parameters $\mathbf{p}'$ is represented by (Doherty et al., 2010):

$$\mathbf{h} = \mathbf{X}\mathbf{p}' + \boldsymbol{\varepsilon} \tag{7}$$

where **ε** is a vector expressing the measurement noise associated with measurements **h**. This assumes that the (linear) model **X** is free from structural defects. Substitution of (7) into (5) yields (Moore and Doherty, 2006):

$$\underline{\mathbf{p}} = \mathbf{G}\mathbf{X}\mathbf{p}' + \mathbf{G}\boldsymbol{\varepsilon} = \mathbf{R}\mathbf{p}' + \mathbf{G}\boldsymbol{\varepsilon} \tag{8}$$

where **R** (= **GX**) is a resolution matrix. Where measurement noise is small enough such that it can be neglected (i.e., the **Gε** term on the right-hand side of (8) equals zero), the **R** matrix describes the relationship between estimated parameter values $\underline{\mathbf{p}}$ and their real-world counterparts $\mathbf{p}'$. Each row of **R** contains the weights by which real-world parameter values are multiplied to produce a particular estimated parameter (commonly referred to as "integration kernel"; Vasco et al., 1997; Paradis et al., 2015). Note that $\underline{\mathbf{p}}$ and $\mathbf{p}'$ are equivalent if **R** equals **I** (and **ε** equals zero). The identifiability of parameter i (d_i) is defined generally as the ability of model calibration to constrain parameter i, and is given by (Doherty and Hunt, 2009):

$$d_i = [\mathbf{R}]_{i,i} \tag{9}$$

where subscript i,i designates the ith diagonal element of **R**. The value of d_i can vary between zero (indicating that the parameter in question is completely unidentifiable) and one (indicating that the parameter in question is completely identifiable). Note that a d_i value of one does not necessarily mean that its estimated value will be error-free, due to the effects of measurement uncertainty. However, statistics such as parameter error variance (e.g., Moore and Doherty, 2005) are not reported in this current study because measurement uncertainty is assumed to be negligible (given the model-generated nature of the synthetic calibration dataset). To explore the influence of model non-linearity, two Jacobian matrices (which are required for the calculation of **R** and **G**) are considered based on both initial and calibrated parameter values.

3.5. Inverse modelling cases

Heads generated on the basis of the reference model are used as "observations" for the calibration dataset in subsequent calibration cases. Heads are taken at pilot point locations where the QL and TS sediments are saturated, resulting in a total of 217 head observations derived from the reference model. The use of coincident observations and estimable parameters facilitates the comparison of reference model- and calibration-based recharge and K distributions at discrete locations (i.e., at pilot points), without the complication of kriging effects (i.e., intra-pilot point recharge and K variability). Head observations are weighted uniformly.

Parameter value constraints in the form of upper and lower bounds, and initial values are given in Table 2. Constraints on hydraulic parameters are consistent with those used for the reference model. K and C parameters are log-transformed within the calibration process. The influence of initial K and C values on estimated values is explored using an alternative set of initial values equal to reference-model parameters.

The spatial variability of recharge is achieved using pilot point parameters co-located with those used for the estimation of K. An isotropic variogram of exponential type is used to produce cell-by-cell recharge variability, as is the case for K variability.

A spatially uniform recharge rate of 100 mm/y, which is equal to the spatially averaged (and temporally averaged over the pre-development period) value of Ordens (2014), is prescribed as an initial value (i.e., prior to calibration). This allows the degree to which the spatial distribution of recharge can be inferred through calibration to be investigated systematically. Spatially uniform recharge rates of 80 and 120 mm/y are also considered to explore the effect of initial recharge values. For the contemporary steady-state model, a spatially uniform recharge rate of 69 mm/y is used, which is equal to the spatially averaged value of Ordens (2014) during the contemporary period.

Because the reference-model recharge distribution varies on a cell-by-cell basis compared to calibrated recharge distributions, which are based on kriging of pilot-point values, unavoidable discrepancies exist between these distributions. For the purpose of generating a reference recharge distribution that is directly comparable to those obtained through calibration, a smoothed representation of the reference recharge distribution is produced by kriging pilot-point values (using a consistent variogram) in such a way that the difference between the original reference distribution and its kriged variant are minimised in a least-squares sense. The reader is referred to Moore (2006) for a detailed description of

Table 2
Parameterisation set-up for inverse modelling cases.

Model parameters/ stresses	Parameterisation method (number of parameters)	Bounds	Initial values[a]
Recharge [mm/y]	Pilot points (125)	0–300	100[b]
K [m/d]			
QL	Pilot points (125)	1–10^4	100
TS	Pilot points (125)	0.132–1320	13.2
C [m^2/d][c]			
UE (TS)	Single value	0.36–3600	36
UW1 (TS)	Single value	0.054–540	5.4
UW2 (TS)	Single value	0.059–590	5.9
UW3 (TS)	Single value	0.23–2300	23
CB1 (TS)	Single value	0.29–2900	29
CB2 (TS)	Single value	0.26–2600	26
CB3 (QL)	Single value	0.18–1800	18
SE (TS)	Single value	0.07–770	7.7

[a] Alternative initial recharge rates of 80 and 120 mm/y, and initial hydraulic parameter values from the reference model are used to assess the influence of initial parameter values.
[b] A recharge rate of 69 mm/y is used for the contemporary steady-state model.
[c] Acronyms relate to the linkages through which USB is connected to adjacent basins (see Knowling et al. (2015) for more details): "UE" refers to Uley East, "UW" refers to Uley Wanilla, "CB" refers to Coffin Bay, and "SE" refers to south-east.

this approach. Fig. 2 shows the smoothed representation of the reference recharge distribution.

Three inverse modelling cases are considered initially, as summarised in Table 3. The first case ("Case 1") involves the estimation of recharge subject to fixed hydraulic parameter values from the reference model. Regularisation is not used for Case 1, and hence, the calibration process involves the minimisation of Φ_m only. Case 1 represents a best-case scenario given that all K and C parameter values are known. Although Case 1 constitutes an unrealistic situation in practice (i.e., an excessive degree of parameter knowledge), it provides a framework for demonstrative and comparative purposes.

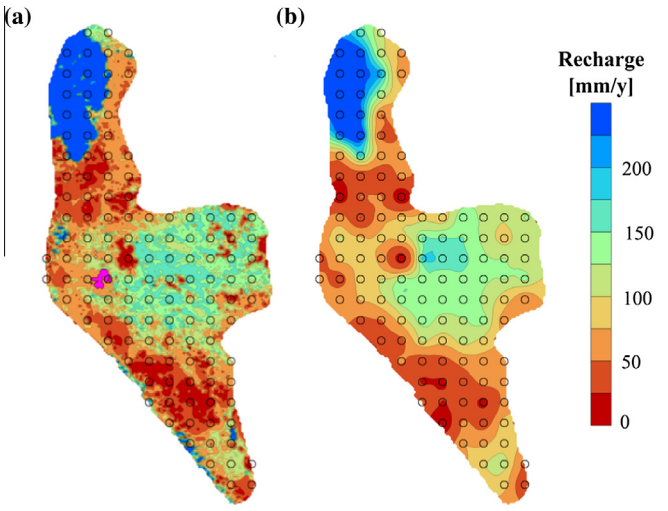


Fig. 2. (a) Reference-model recharge distribution, and (b) its counterpart obtained from kriging pilot-point values that are calculated such that cell-by-cell recharge differences with (a) are minimised. The circles depict pilot point locations.

Case 2 involves the estimation of both recharge and hydraulic parameters in the absence of regularisation constraints (Table 3). Case 2 represents a worst-case scenario, whereby no expert knowledge regarding K and C is available. Because the Case 2 calibration involves correlated parameters that are largely unconstrained, it is expected that considerable parameter compensation will occur. Nevertheless, this case serves as a useful demonstration and for comparing subsequent cases.

In Case 3, the estimation of recharge and hydraulic parameters is subject to varying degrees of hydraulic parameter information, imparted by regularisation. The Case 3 calibration therefore involves the minimisation of both Φ_m and Φ_r. The Φ_m^l is specified such that a relatively comparable level of model-to-reference model misfit is obtained for all cases. This is achieved by prescribing a value for Φ_m^l that slightly exceeds Φ_m obtained for Cases 1 and 2 given that regularisation is expected to produce higher Φ_m values (due to the minimisation of both Φ_m and Φ_r).

Case 3 includes eleven variants (Cases 3a to 3k), which vary according to the number of Tikhonov regularisation constraints imposed. The first five cases (3a to 3e) employ preferred K and C values, which accord with the reference model. The number of preferred values is based on different percentages of the total number of calibrated parameters (i.e., 258 parameters: 125 QL K, 125 TS K, and eight C), as follows: 10% (25 K and one C); 25% (62 K and two C); 50% (125 K and four C); 75% (188 K and six C); and 90% (225 K and seven C). Fig. 3 shows the location of preferred K and C values for each sub-case. Uniform weights are assigned to all preferred parameter values. Cases 3f to 3j are the same as Cases 3a to 3e except with the additional constraint of preferred homogeneity in K. The use of preferred K-homogeneity is considered reasonable here given that the generation of reference-model K distributions involved preferred K-homogeneity. An additional case (3k) is considered whereby only preferred K-homogeneity is employed (i.e., no preferred values). Weights on preferred homogeneity

Table 3
Inverse modelling cases.

ID	Estimable parameters	Description	Hydraulic parameter information	Regularisation used?
Case 1	Recharge	Perfect knowledge of hydraulic parameters	K and C fixed as per reference model	No
Case 2	Recharge, K, C	No prior knowledge of hydraulic parameters	Parameter bounds only	No
Cases 3a to 3k	Recharge, K, C	Varying degree of knowledge regarding hydraulic parameters	Preferred K and C values (from reference model); preferred K-homogeneity	Yes

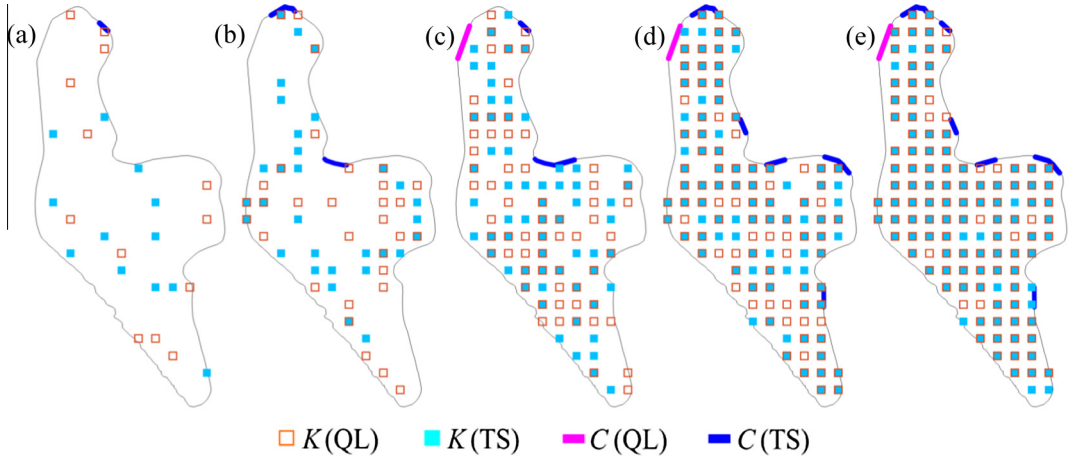


Fig. 3. Distribution of preferred K and C values used for Cases 3a to 3e (and for Cases 3f to 3j). The number of preferred values is based on the following percentages of the total number of adjustable hydraulic parameters: (a) 10%; (b) 25%; (c) 50%; (d) 75%; and (e) 90%.

Table 4
Model-to-reference model misfit statistics for all inverse modelling cases.

		Additional inverse modelling cases				
	Original Φ_m [m^2] (RMSE [m])	Use of SGD calibration target Φ_m [m^2][a] (RMSE [m], [m^3/d])[b]	Contemporary model Φ_m [m^2] (RMSE [m])	Initial recharge of 80 mm/y Φ_m [m^2] (RMSE [m])	Initial recharge of 120 mm/y Φ_m [m^2] (RMSE [m])	Initial K, C from reference model Φ_m [m^2] (RMSE [m])
Case 1	1.36 (0.008)	1.73 (0.009, 0.9)	4.2 (0.014)	0.12 (0.002)	0.11 (0.002)	–
Case 2	0.94 (0.007)	23 (0.032, 0.6)	0.58 (0.005)	6.2 (0.017)	0.68 (0.006)	0.12 (0.002)
Case 3a	136 (0.079)[c]	15 (0.026, 5.0)	262 (0.110)[c]	2.2 (0.010)	102 (0.068)[c]	1.4 (0.008)
Case 3b	1.9 (0.009)	5.8 (0.016, 1.4)	2.2 (0.010)	2.2 (0.010)	2.4 (0.010)	0.58 (0.005)
Case 3c	2.3 (0.010)	9.3 (0.021, 0.3)	3.9 (0.013)	2.2 (0.010)	2.2 (0.010)	2.2 (0.010)
Case 3d	2.2 (0.010)	2.2 (0.010, 0.1)	6.1 (0.017)	4.5 (0.014)	2.3 (0.010)	2.0 (0.009)
Case 3e	2.2 (0.010)	1.9 (0.009, 1.4)	3.6 (0.013)	2.2 (0.010)	2.1 (0.010)	1.8 (0.009)
Case 3f	2.3 (0.010)	2.5 (0.011, 0.1)	4.7 (0.015)	2.3 (0.010)	2.4 (0.010)	2.3 (0.010)
Case 3g	2.4 (0.011)	3.2 (0.012, 1.2)	2.8 (0.012)	2.3 (0.010)	3.3 (0.012)	2.3 (0.010)
Case 3h	3.3 (0.012)	7711 (0.596, 31)[c]	112 (0.072)[c]	420 (0.139)[c]	510 (0.153)[c]	3.1 (0.012)
Case 3i	5.9 (0.016)	9.0 (0.020, 0.1)	2.6 (0.011)	280 (0.113)[c]	9.1 (0.021)	2.7 (0.011)
Case 3j	2.2 (0.010)	2.3 (0.010, 0.4)	156 (0.085)[c]	2.2 (0.010)	2.2 (0.010)	2.5 (0.011)
Case 3k	285 (0.115)[c]	13,804 (0.799, 100)[c]	5047 (0.482)[c]	7045 (0.570)[c]	13,958 (0.802)[c]	43 (0.045)[c]

[a] Contribution to Φ_m from the model-to-reference model head misfit only.
[b] RMSE for the model-to-reference model head misfit and flux misfit, respectively.
[c] Level of fit considered to be inadequate for the purpose of comparing recharge estimations.

constraints are also uniform, and are equivalent to those assigned to preferred values.

Each of the cases described above are re-evaluated to consider an observation of total submarine groundwater discharge (SGD) to assess whether or not the inclusion of a groundwater-flux calibration target allows for an improved estimation of recharge. The "observed" SGD (2.698×10^7 m^3/y) is obtained from the reference model. A relative observation weight of 0.01 is prescribed to account for differences between flux and head units. All cases are also re-evaluated in terms of recharge estimability for the contemporary model (which includes pumping), and using different initial recharge and hydraulic parameter values for the pre-development model.

4. Results

4.1. Case 1

Table 4 lists model-to-reference model misfit statistics (including values of Φ_m and the corresponding calibration root-mean-squared-error (RMSE)) for all inverse modelling cases. The level of fit obtained for Case 1 (Φ_m of 1.36 m^2, RMSE of 0.008 m) is considered adequate.

Fig. 4 shows the calibrated recharge ($\underline{R}$) values, and their comparison to the kriged reference model recharge (R'_{krig}) values, for Case 1. A reasonable correlation is evident between pilot point $\underline{R}$ and R'_{krig} values (Fig. 4a), as evidenced by a regression line-of-best-fit slope and coefficient of determination (r^2) equal to 0.96 and 0.66, respectively. The regression-line slope approaching unity indicates reasonable agreement between $\underline{R}$ and R'_{krig}, on average, whereas the mid-range r^2 value suggests a moderate degree of scatter. The error in $\underline{R}$ (i.e., $\underline{R} - R'_{krig}$) at each pilot point varies from −139 to 171 mm/y, with an average error and average absolute error of 11 mm/y (model overestimation) and 34 mm/y, respectively. The volumetric $\underline{R}$ error percentage is −1.1% (a small model underestimation). The inconsistency between the average pilot point $\underline{R}$ error (overestimation) and volumetric $\underline{R}$ error (underestimation) is a consequence of kriging, as evidenced by the average $\underline{R}$ error obtained on a model cell-by-cell basis (−0.02 mm/y).

The reasonable agreement between $\underline{R}$ and R'_{krig} is further illustrated by the similarity of their distributions (compare Fig. 4b to Fig. 2b). The $\underline{R}$ field largely resembles the R'_{krig} distribution. The spatial distribution of $\underline{R}$ errors (Fig. 4c) displays complicated patterns that are not attributable to physical factors related to recharge.

Fig. 5 shows the identifiability of recharge parameters employed in Case 1. Identifiability values are highly variable (ranging between zero and one). Just 23 of the 125 parameters are considered to be identifiable on the basis of the calibration dataset, indicated by identifiability values of >0.8. Of the remaining parameters, 44 display mid-range values (0.2–0.8), and 58 display low values (<0.2).

4.2. Case 2

The model-to-reference model misfit of Case 2 (Φ_m of 0.94 m^2, RMSE of 0.007 m) is smaller than that of Case 1. This occurs because, in the absence of regularisation, the calibration has greater freedom to modify parameters relative to Case 1. Case 2 represents the best fit to head observations relative to other cases that exclude flux data and that adopt an initial recharge value of 100 mm/y (Table 4).

Fig. 6 shows $\underline{R}$ values, and their comparison to R'_{krig} values, for Case 2. A correlation between $\underline{R}$ and R'_{krig} is lacking, as evidenced

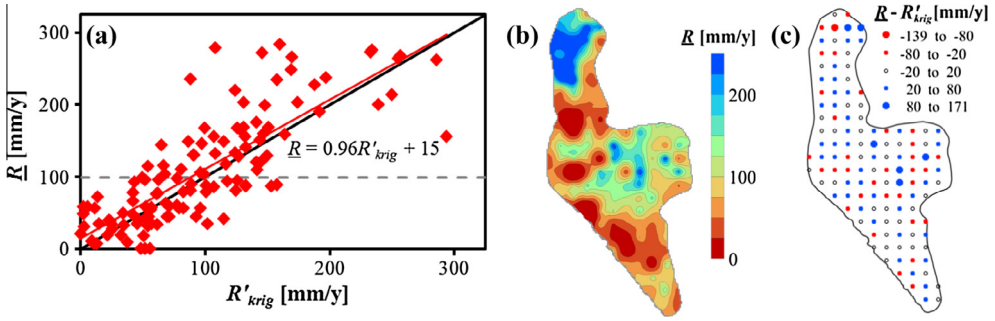


Fig. 4. (a) $\underline{R}$-versus-R'_{krig} scatterplot (the black line represents the 1:1 line, the red line represents the regression line-of-best-fit, as given by the equation shown, and the dashed line represents the initial uniform $\underline{R}$ value of 100 mm/y), (b) $\underline{R}$ distribution [mm/y] and (c) $\underline{R}$ error distribution [mm/y], for Case 1.

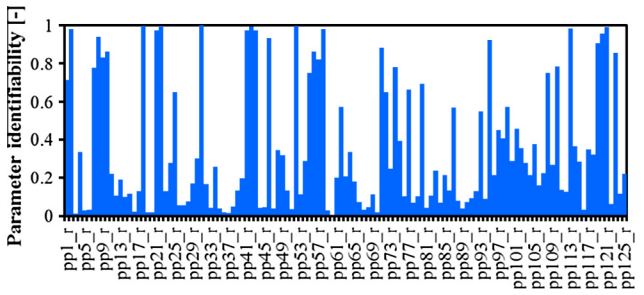


Fig. 5. Recharge parameter identifiability for Case 1. The x-axis labels are names assigned to the 125 pilot points.

by the regression-line slope of 0.06 (Fig. 6a). The majority of $\underline{R}$ values are only marginally modified by calibration (67% of values lie between 80 and 120 mm/y). The average error and average absolute error of $\underline{R}$ at each pilot point is −14 mm/y (model underestimation) and 54 mm/y, respectively. The volumetric $\underline{R}$ error percentage is −15%, which is consistent with the average cell-by-cell-based error (−15 mm/y).

The $\underline{R}$ distribution (Fig. 6b) does not display consistent patterns compared to those of R'_{krig} (Fig. 2b). For example, the high recharge rates associated with the north-west (>200 mm/y) and central (>150 mm/y) regions of USB are not reflected in the $\underline{R}$ distribution. These inconsistencies are a result of the compensatory roles assumed by hydraulic parameters, i.e., spurious hydraulic parameter values are obtained to compensate for errors in $\underline{R}$ to maintain an equivalent level of model-to-reference model fit. For example, in the aforementioned areas where $\underline{R}$ is less than R'_{krig}, calibrated QL K values are lower than those from the reference model, as evidenced by negative K (QL) errors (Fig. 6c). The average $\underline{R}$ error of −15 mm/y is compensated by an average pilot-point K (QL) error of −472 m/d.

Fig. 7 shows the identifiability of hydraulic and recharge parameters employed in Case 2. While many of the hydraulic parameters display mid-range identifiability values (Fig. 7a–c),

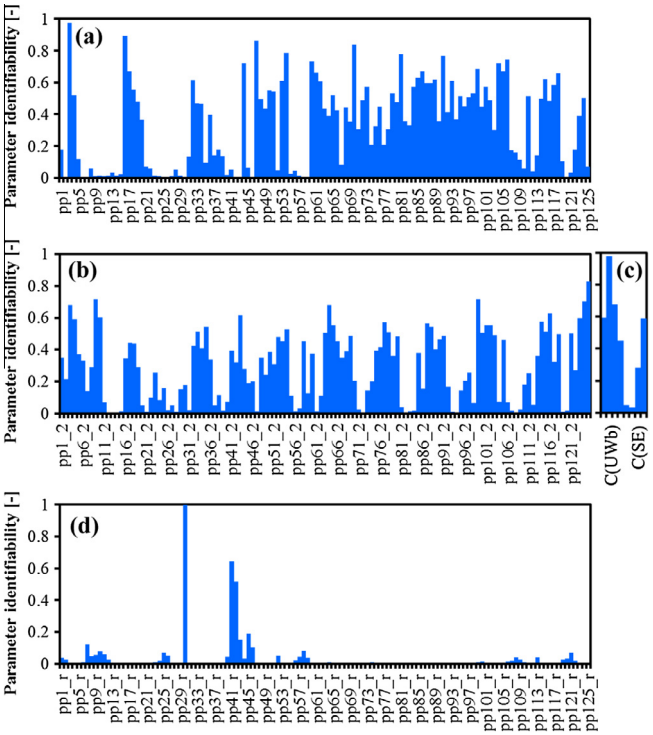


Fig. 7. Identifiability of (a) QL K, (b) TS K, (c) C, and (d) recharge parameters for Case 2. The x-axis labels are names assigned to the pilot point parameters.

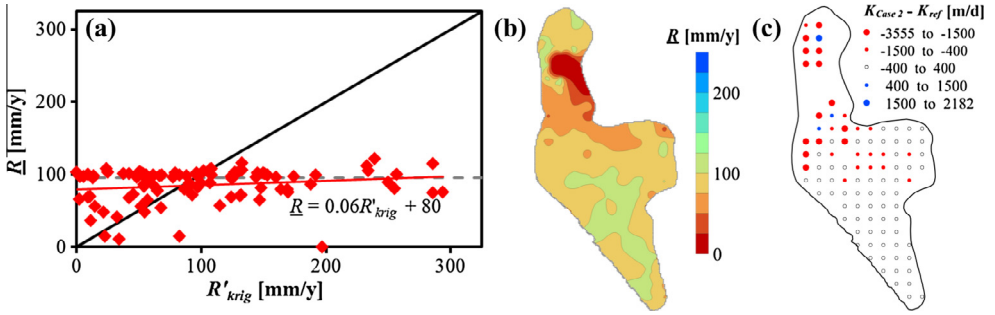


Fig. 6. (a) $\underline{R}$-versus-R'_{krig} scatterplot, (b) $\underline{R}$ distribution [mm/y], and (c) K (QL) error distribution [m/d], for Case 2. In (c), the K-error is not displayed at some pilot points because the simulated head in the QL aquifer is below the cell-bottom elevation.

only seven of the 270 hydraulic parameters are deemed identifiable (indicated by values of >0.8). The identifiability of all but three recharge parameters (which display values of 1.0, 0.64, 0.52) is low (<0.2), with the majority of parameters displaying values of approximately zero (Fig. 7d).

4.3. Case 3

A Φ_m^l value of 2.17 m^2 is specified for Case 3 such that the corresponding calibration RMSE value (0.01 m) slightly exceeds that obtained for Cases 1 and 2. The model-to-reference model misfit for seven of the eleven cases (3b, c, d, e, f, g and j) is within ~15% of the target (Φ_m values < 2.50 m^2, RMSE values <0.0106 m) (Table 4), therefore allowing for reliable comparisons of recharge estimates to be made between the Case 3 variants. The levels of misfit for Cases 3 h and 3i are higher (Φ_m values of 3.3 and 5.9 m^2, RMSE values of 0.012 and 0.016 m, respectively), but nonetheless are considered sufficient for comparative purposes. Acceptable calibration results, in terms of Φ_m, could not be obtained for Cases 3a and 3k. Higher Φ_m values are attributable to the role of regularisation within the calibration process, whereby Φ_m is penalised by the requirement to minimise Φ_r.

Table 5 lists average deviations from preferred *K* and *C* values, and average differences between neighbouring pilot-point *K* values. Generally, the deviation from preferred hydraulic parameter values for Cases 3a to 3g is considered insignificant. For Cases 3f to 3i, the extent to which estimated hydraulic parameter values deviate from preferred values generally increases due to the introduction of preferred homogeneity. This occurs because the preferred smoothness counteracts the ability of inverse modelling to reproduce the preferred values of *K*. The average difference between neighbouring pilot-point *K* values for Cases 3f to 3k is considered to be relatively small.

Fig. 8 shows the $\underline{R}$ values, in comparison to R'_{krig} values, for Case 3a, c, e, f, h, j and k (other cases are omitted for brevity). The degree to which $\underline{R}$ and R'_{krig} are correlated increases from Case 3a to 3e and from Case 3f to 3j, i.e., with the number of preferred value constraints, as evidenced by the regression-line slopes given in Fig. 8. A higher variability in $\underline{R}$ values is apparent for Cases 3f, h and j compared to 3a, c and e, as indicated by larger coefficients of variation (*CV*; i.e., ratio of standard deviation to mean) values (0.21, 0.44, and 0.58 compared to 0.11, 0.22, and 0.38, respectively).

Fig. 9a and b shows the average error and average absolute error of $\underline{R}$ estimates (based on cell-by-cell values) for all cases. Cell-by-cell recharge values are assessed given that average errors based on these values display consistency with volumetric recharge error. In general terms, a reduction in $\underline{R}$ error is evident as the number of preferred hydraulic parameter values increases, as expected given that preferred values guide the calibration process closer to the parameter set of the reference case. A clear error reduction trend is apparent between Cases 3b and 3e, and Cases 3f and 3k. Inconsistencies in this trend are evident for Case 3a, occurring as a result of a less-than-sufficient calibration fit. For cases where an adequate fit is obtained (all except 3a and k), the average error is 3% smaller. The average absolute error is 18% smaller, on average, with preferred *K*-homogeneity. Although errors based on Cases 1 and 2 cannot be compared directly to those

Table 5
Average deviation from preferred *K* values [m/d] and *C* values [m^2/d], and average difference between neighbouring pilot-point *K* values (for cases involving preferred homogeneity only; denoted "*K*-u" below) [m/d] for Cases 3a to 3k.

	Original			Additional inverse modelling cases													
				Use of SGD calibration target			Contemporary model			Initial recharge of 80 mm/y			Initial recharge of 120 mm/y			Initial *K*, *C* from reference model		
	K	*C*	*K*-u	*K*	*C*	*K*-u	*K*	*C*	*K*-u	*K*	*C*	*K* – u	*K*	*C*	*K* – u	*K*	*C*	*K* – u
Case 3a	0.73	0.008	–	31	1.2	–	1.4	0.001	–	0.003	0	–	41	0.76	–	0.004	0	–
Case 3b	0.003	0	–	83	0.003	–	0.001	0	–	0	0	–	0.007	0	–	0	0	–
Case 3c	0.004	0	–	133	4.2	–	16	0.71	–	0	0	–	0	0	–	0.002	0	–
Case 3d	0.068	0.001	–	72	0.66	–	3.9	0.42	–	32	0.26	–	0.014	0	–	0.004	0	–
Case 3e	0.044	0	–	86	1.1	–	1.3	0.03	–	0.005	0	–	0.003	0	–	0	0	–
Case 3f	0.12	0	1.8	0.19	0.001	1.9	0.74	0	4.6	0.14	0	1.8	0.12	0	1.8	0.042	0	1.8
Case 3g	0.30	0	1.7	12	0	1.8	1.4	0.01	5.3	0.31	0	1.7	0.34	0	1.7	0.22	0	1.7
Case 3h	2.0	0.008	1.7	2.3	0	1.5	6.6	0.27	3.9	2.2	0.002	1.6	3.2	0.002	1.6	1.3	0.001	1.7
Case 3i	7.5	0.21	1.7	12	0.22	1.8	13	0.03	3.0	10	0.56	1.2	19	0.80	1.2	4.0	0.009	1.2
Case 3j	6.1	0.005	1.7	62	0.44	1.8	18	0.28	1.8	6.0	0.002	1.7	6.2	0.006	1.7	6.5	0.004	1.7
Case 3k	–	–	1.4	–	–	1.1	–	–	1.0	–	–	1.1	–	–	1.2	–	–	1.6

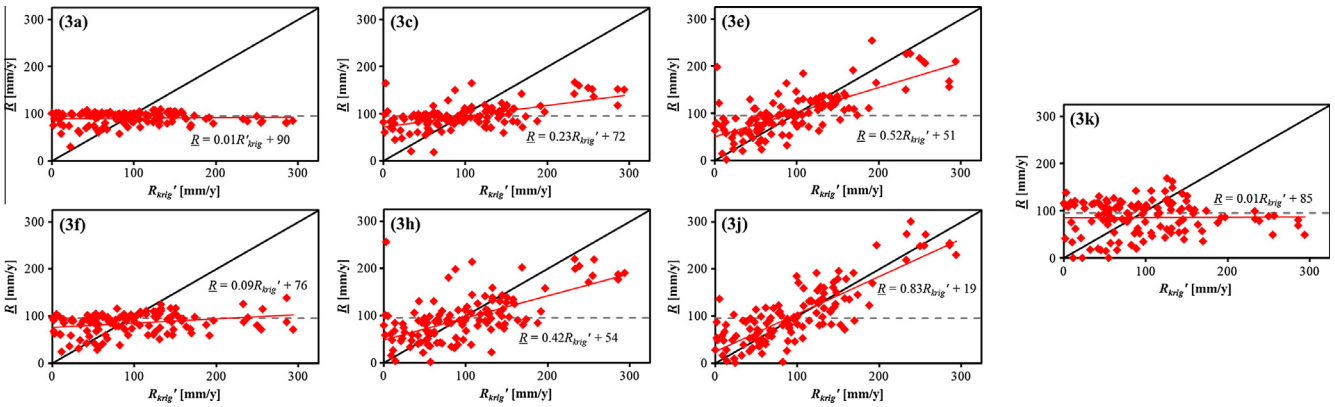


Fig. 8. $\underline{R}$-versus-R'_{krig} scatterplots for Case 3a, c, e, f, h, j and k.

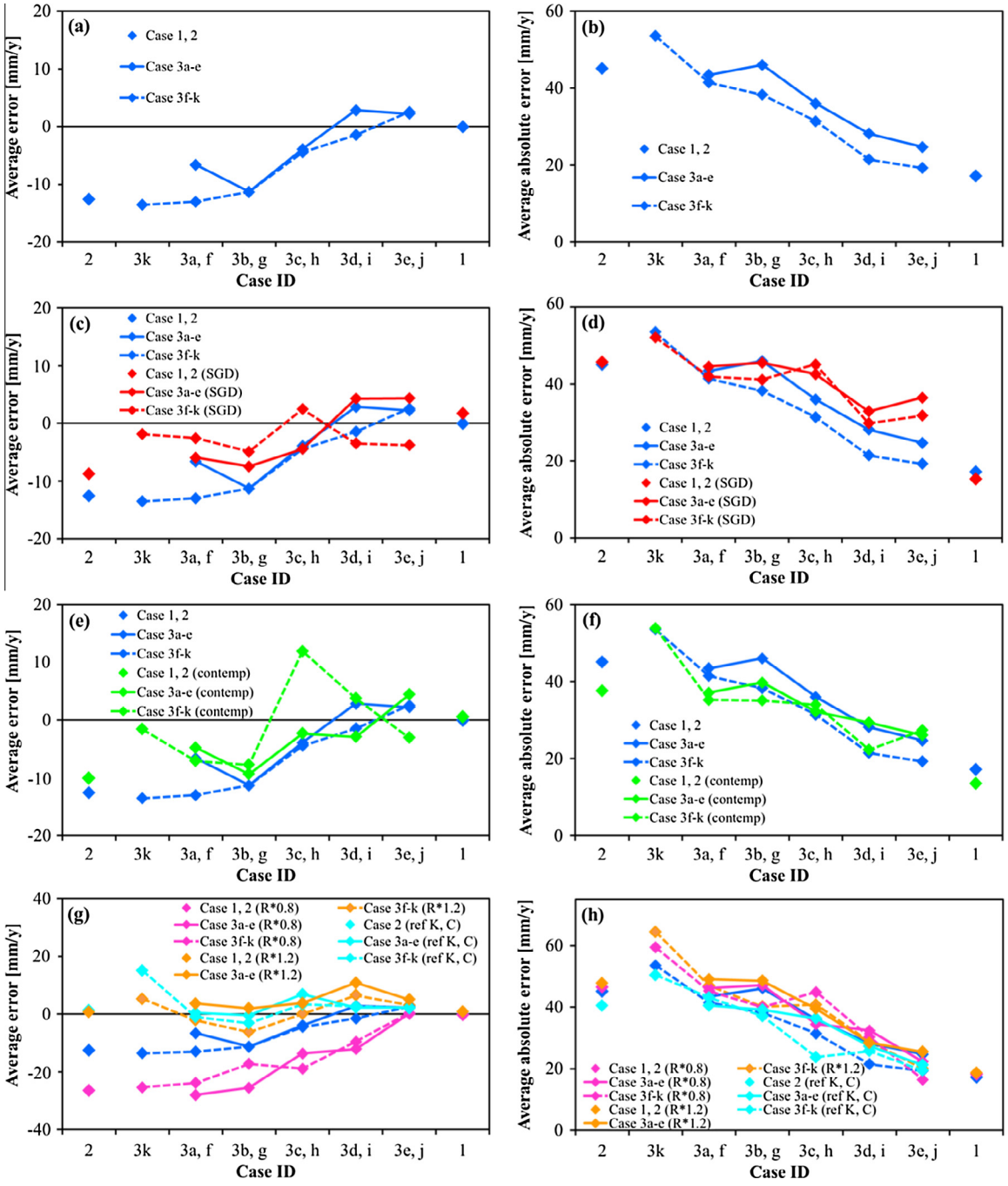


Fig. 9. Average error and average absolute error of $\underline{R}$ for (a and b) Cases 1 to 3, (c and d) Cases 1 to 3 with a SGD calibration target, (e and f) Cases 1 to 3 for the contemporary steady-state model ("contemp"), and (g and h) Cases 1 to 3 with initial recharge values of 80 mm/y ("R*0.8") and 120 mm/y ("R*1.2"), and initial hydraulic parameter values equal to those of the reference-model ("ref K, C"). Errors are based on cell-by-cell recharge values. The blue markers and lines in (g) and (h) are results from the original cases shown in (a) and (b). Note the different y-axis scale on (g). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of Case 3 (given the difference in the calibration approaches), they display consistent patterns in error reduction with increasing degrees of hydraulic parameter information.

4.4. Use of SGD as a calibration target

The thirteen inverse modelling cases (1 to 3k) presented above are re-evaluated with the addition of an SGD calibration target. The model-to-measurement misfit for eleven of the thirteen cases (1, 2, 3a, b, c, d, e, f, g, i and j) is considered to be sufficiently low to allow for reliable comparisons of recharge estimates between cases (RMSE values of <0.032 m (for head) and <5 m^3/d (for flux); Table 4). The deviation from both preferred K and C values is considered large for all cases except 3f (Table 5). Cases 3a, b, c, d, e and j, in particular, display very large deviations from preferred K values. This highlights that the inclusion of the flux target produces generally worse estimates of spatially distributed hydraulic parameters (and therefore recharge, as discussed below and in Section 5) due to the addition of SGD to the calibration objective function.

Fig. 9c and d shows that a reduction in $\underline{R}$ error is generally apparent with increasing number of preferred hydraulic parameter values, with or without the flux target. Errors for Cases 3a, e, h and j do not display consistent patterns as a result of either a less-than-sufficient model fit (for Case 3h), or large deviations from preferred values (for Cases 3b, e and j). The amount by which the error is reduced through addition of preferred values, however, is smaller where the flux target is applied. In particular, the average recharge error is not reduced from Case 3f to 3j. This is at least partly due to the smaller average errors obtained where fewer preferred values are used (Cases 3a, b, f and g) (note the inadequate level of fit obtained for Case 3a without the flux target, therefore precluding a reliable comparison between Case 3a with and without the flux target). Of the ten cases where a reliable comparison can be made between cases with and without the inclusion of the flux target (i.e., where sufficient model fits are obtained; all cases except 3a, h and k), the average error is smaller for cases 2, 3b, f and g (the average error is 25% smaller, on average, across the ten cases), and the average absolute error is smaller for cases 1 and 3b (the average error is 14% larger, on average, across the ten cases) where the flux target is used.

4.5. Application of an alternative steady-state condition

Inverse modelling cases 1 to 3k are re-evaluated for the contemporary steady-state model. The model-to-reference model misfit for nine of the thirteen cases (1, 2, 3b, c, d, e, f, g, and i) is considered adequate for the purpose of comparing recharge estimates between cases (RMSE values <0.017 m; Table 4). The deviation from both preferred K and C values is considered insignificant for Case 3b only (Table 5). The difference between neighbouring K values, on average, for Cases 3f to 3j is relatively small (Table 5).

Fig. 9e and f shows the (cell-by-cell-based) average error and average absolute error of $\underline{R}$ for all cases involving the contemporary steady-state model. A reduction in $\underline{R}$ error is apparent with increasing number of preferred hydraulic parameter values, similar to the error trends identified from the pre-development model. The error-reduction trend is particularly apparent for Case 3b to 3e and 3g to 3j (note that, while the average error of Cases 3g and 3h are of opposite sign, the average error magnitudes and the average absolute errors are comparable). For cases where a sufficient model fit is obtained for both the pre-development and contemporary models (all except 3a, h, j and k), the average error is 18% smaller, and the average absolute error is 9% smaller, on average, where the contemporary model is used.

4.6. Effect of initial recharge and hydraulic parameter values

Cases 1 to 3k are re-evaluated using different initial recharge and hydraulic parameter values. The model-to-reference model misfit for 31 of the 38 cases (thirteen cases for an initial recharge rate of 80 mm/y, thirteen cases for an initial recharge rate of 120 mm/y, and twelve cases for initial hydraulic parameter values specified as reference-model values) is considered adequate for comparing recharge estimates between different cases (RMSE values <0.021 m; Table 4). The deviation from preferred hydraulic parameter values is large only for Cases 3h to 3j for each of the different initial parameter value sets (Table 5). For Cases 3f to 3j, the deviation from preferred K values generally increases (Table 5) because the imposition of preferred K-homogeneity counteracts that of preferred values.

Fig. 9g and h shows the (cell-by-cell-based) average error and average absolute error of $\underline{R}$ for cases with different initial recharge and hydraulic parameter values. The error-reduction trend (with increasing number of preferred hydraulic parameter values) is generally apparent for all cases. Average $\underline{R}$ values are larger where the initial recharge value of 120 mm/y is applied, and smaller where the initial recharge value of 80 mm/y is applied, compared to the initial recharge value of 100 mm/y (Fig. 9g). Relative to the original cases, the average error is 100% larger, on average, with an initial recharge value of 80 mm/y (excluding Cases 3a, h, i and k due to inadequate model-fit), and 33% smaller, on average, with an initial recharge value of 120 mm/y (excluding Cases 3a, h and k due to inadequate model-fit). The average absolute error is 2% and 8% larger, on average, for initial recharge values of 80 and 100 mm/y, respectively, compared to the original cases. Both the average error and average absolute error is smaller with the initial hydraulic parameter values from the reference model compared to the original cases (61% and 6% smaller, on average, for all cases except 3a and k, where poor fits are obtained).

5. Discussion

The level of model fit (i.e., in terms of head RMSE) and the accuracy of estimated recharge values displays a poor correlation amongst the various cases. This is evidenced by r^2 values of 0.02 and 0.03 for regression lines representing RMSE-average recharge error and RMSE-average absolute recharge error, respectively (for cases where an adequate model fit is obtained). This is consistent with previous studies (e.g., McKenna et al., 2003; Hill and Tiedeman, 2007) that show that the head match is not a proxy for accurate parameter values. The attainment of inaccurate recharge values despite low model misfit exemplifies the compensatory roles assumed by hydraulic parameters in offsetting errors in recharge, as illustrated by, e.g., the underestimation of K values where recharge values are lower than those of the reference model (Fig. 6c).

Inverse modelling results demonstrate that the ability of the calibration process to inform recharge varies significantly with the degree to which hydraulic parameters are constrained, in agreement with Sanford (2002). Where all hydraulic parameters are known, it is shown that spatially averaged recharge can be estimated accurately (average cell-by-cell error of −0.02%), whereas spatially distributed recharge is less well-constrained, as evidenced by the moderate degree of scatter about the regression line (r^2 value of 0.66; Fig. 4). That is, an accurate basin-scale estimate of recharge does not guarantee that a reasonable recharge distribution has been obtained from model calibration. While failure of the calibration process to identify recharge spatial variability may not impact the model's predictability (e.g., to estimate drawdown) directly, there are important implications for the distribution of K values, which are more likely to influence the prediction of pumping impacts (e.g., Freeze and Cherry, 1979). Where aquifer parameters are known, discrepancies in the calibrated recharge distributions are caused by compensation between recharge parameters, as evidenced by low recharge identifiability values (Fig. 5).

The amount of hydraulic parameter information required to obtain a reasonable estimate of recharge, in particular its spatial variability, on the basis of the current findings, appears to be excessive from a practical perspective. For cases with initial recharge values based on the average reference-model recharge rate and literature-derived initial hydraulic parameter values (Table 2), a reasonable spatially averaged recharge estimate generally requires at least 129 of the 258 reference-model hydraulic parameter values to be set as regularisation constraints (<10% average error is obtained for 23 of the 31 cases where an adequate model fit is obtained; Fig. 9a, c and e). While average recharge can be estimated with a reasonable degree of accuracy for these cases,

the spatial variability of recharge requires higher levels of hydraulic parameter information. This is evidenced by the requirement for at least 232 preferred hydraulic parameter values to achieve average absolute recharge errors of <20% (average absolute error <20% is obtained for 4 of the 31 cases where an adequate model fit is obtained; Fig. 9b, d and f).

The incorporation of preferred *K*-homogeneity is shown to be of benefit to the estimation of recharge for some cases. In particular, the imposition of *K*-homogeneity is shown to produce, on average, improved estimation of average recharge (average error is 14% lower), and spatial variability in recharge (average absolute error is 15% lower), across all cases that involve regularisation where an adequate fit is obtained (total of 18 cases). This is because smoother *K*-fields lead to enhanced spatial variability in recharge, due to the effects of parameter compensation and the heterogeneous nature of the reference model (Fig. 9b). However, the benefit of preferred *K*-homogeneity to recharge estimability for other real-world situations is yet to be demonstrated.

The use of SGD, which can be inferred in practice from environmental tracers and other methods (e.g., Burnett et al., 2006; Hougham et al., 2008), is shown to be of variable worth as a calibration target in the estimation of recharge for the inverse modelling cases considered in the current study. On average, the inclusion of SGD improves the estimation of spatially averaged recharge (25% smaller average error; Fig. 9c), but not the estimation of spatial recharge variability (14% larger average absolute errors; Fig. 9d). The improvement in average recharge estimability occurs because of the relationship between SGD and recharge (Cheng and Ouazar, 1999). Such improvements occur only where relatively few (<129) preferred hydraulic parameter values are employed. This indicates that the constraint on average recharge imparted by having a large number of preferred values distributed over the basin is similar to that imparted by a basin-wide flux target (e.g., SGD). The reduction in the estimability of spatially variable recharge with the addition of the SGD target is consistent with the large deviations from preferred hydraulic parameter values obtained with the SGD target (Table 5). The use of distributed flux data (e.g., spatial variability in SGD), while not considered here, is likely to improve the constraint on the spatial variability in recharge.

Results from the contemporary steady-state model are, in general terms, similar to those obtained from the pre-development steady-state model. This indicates that trends in recharge estimability are not specific to a particular steady-state condition (Fig. 9e and f). The findings of the current investigation are therefore somewhat validated, notwithstanding that both models share many similarities. The simulation of pumping is shown to produce, on average, lower recharge errors than the pre-development model (average error is 18% lower and average absolute error is 9% lower), given that the prescribed flux provides an additional constraint to the inverse problem (Zhang, 2014). In general terms, the results from both the addition of SGD as a calibration target and pumping suggest that the ability of groundwater fluxes to constrain the estimation of recharge is complicated and problem-dependent. This is supported by Voss (2011), who advised that the incorporation of alternative data types (i.e., besides head) to assist in constraining the uncertainty of groundwater model parameters may not always be as effective as commonly thought.

The degree to which hydraulic parameters must be constrained to achieve reasonable recharge estimates is shown to be highly dependent on initial recharge and hydraulic parameter values (e.g., <10% average error is obtained for initial recharge values of 120, 100 and 80 mm/y using 0%, 50% and 90% of reference model-based preferred values, respectively; Fig. 9g). The role of initial parameter values in constraining the calibration process is highlighted by the larger average recharge error where a larger initial recharge value is specified (120 mm/y), compared to the smaller average recharge error where a smaller initial recharge value is specified (80 mm/y) (Fig. 9g). This is in agreement with previous studies that demonstrate the importance of initial parameter values within traditional groundwater model calibration problems (e.g., Bravo et al., 2002; Bahremand and De Smedt, 2008). Differences in estimated parameter values due to different initial values occur because of the presence of multiple objective function minima (e.g., Hill and Tiedeman, 2007).

The analysis of parameter identifiability provides insight into the degree of non-uniqueness between parameters of both the same type and different types on a spatially distributed basis. For example, significantly lower recharge identifiability is evident where hydraulic parameters are also estimated as part of the calibration process (110 of the 125 recharge identifiability values <0.05; Fig. 7c) compared to where hydraulic parameters are known (22 of the 125 recharge identifiability values <0.05; Fig. 5). This indicates that the ability of the calibration process to reduce the potential for wrongness in recharge estimates is small, due to the need to jointly estimate hydraulic parameters on the basis of the same dataset. The spatial variability in identifiability values display complicated patterns that are related primarily to the proximity of boundary conditions (e.g., highest *K*-identifiabilities are located adjacent to the fixed-head ocean boundary) and the extent of aquifers (e.g., highest recharge-identifiabilities are located where recharge occurs within the TS aquifer, underlying unsaturated QL sediments). The distribution of observation data typically dominates such patterns (e.g., Fienen et al., 2010; Wallis et al., 2014); this is not the case here because of the evenly distributed head targets. Confidence in these results is enhanced given that the effect of model non-linearity on identifiability values is relatively minor (values differ by 8% on average). A strong inverse correlation between parameter identifiability and post-calibration parameter error variance (r^2 values of >0.99; not shown for brevity) is obtained given the assumption of negligible measurement errors, which is considered reasonable for a synthetic reference model (Section 3.4). Therefore, the identifiability values reported in this study can be considered as a proxy for post-calibration parameter uncertainty (e.g., Moore and Doherty, 2005).

In this study, the ability of the calibration process to inform parameters is evaluated primarily with respect to the error associated with parameter estimates, which can be quantified through the application of a reference model. In practice, however, estimated parameter values cannot be evaluated in terms of error. The reasonableness of parameter values must therefore be assessed in terms of whether or not they can be considered plausible on the basis of field data such as aquifer test-derived *K* estimates (e.g., Langevin and Zygnerski, 2013; Knowling et al., 2015). Such assessments are limited by the need to consider the scale-dependency of hydraulic properties (e.g., Sánchez-Vila et al., 2006; Li et al., 2011).

In studies that apply unsaturated zone models to estimate recharge, the parameters of recharge models are constrained using expert knowledge related to, e.g., land-surface and unsaturated-zone processes (Jyrkama et al., 2002; Chemingui et al., 2015). In the current study, recharge is estimated solely through the application of the regional-scale groundwater model, and recharge modelling serves merely to provide the reference-case recharge spatial variability. We focus on aspects of the groundwater model, aside from expert knowledge of recharge itself, in attempting to inform recharge independent of other recharge estimation techniques. Further work, including coupling of the recharge model with MODFLOW, is required to investigate the extent to which recharge can be informed by groundwater model calibration with varying degrees of information pertaining to both recharge

(e.g., spatial zones of land use, soil type, etc.) and aquifer hydraulic properties.

While this study shows that the estimation of recharge through calibration of field-scale steady-state groundwater models may be limited, the estimability of recharge through calibration of transient groundwater models, which involves considerably more observations than steady-state models, yet requires the additional consideration of storage parameter uncertainty, remains unknown for practical situations. The degree to which field-scale transient model calibration can inform recharge and its variability in time is the subject of future work.

It is expected that the findings of this study are to some degree dependent on the unique characteristics of the USB reference model (e.g., the lack of groundwater-dependent streams). Application of similar analyses to other field sites, for example, temperate climates where surface water and groundwater are often hydraulically connected, may reveal an enhanced estimability of recharge due to the availability of stream discharge data, which may reduce parameter non-uniqueness. Therefore, the investigation of recharge estimability for other field sites is warranted to extend the current findings to a wider range of situations, in particular where measurements of groundwater flux are more readily attainable than is possible for USB.

6. Conclusions

Non-uniqueness poses one of the primary challenges to the estimation of groundwater model parameters. The current study focuses on the ramifications of non-uniqueness between recharge and aquifer hydraulic parameter values in terms of recharge estimability through groundwater model calibration. Presently, the degree to which recharge can be informed through calibration is largely unknown for complex field-scale situations. This study evaluates the ability of calibration to inform recharge within a regional setting using a series of highly parameterised steady-state inverse modelling experiments containing varying degrees of hydraulic parameter constraints.

Inverse modelling results show that the ability of the calibration process to inform recharge varies significantly with the degree to which hydraulic parameters are constrained. For the field-scale groundwater model considered, reasonable estimates of recharge are shown to require large amounts of hydraulic parameter information. For example, an average error of <10% requires that 50% of the reference-model hydraulic parameters are included in the calibration using preferred-value regularisation (>100 K values across the 129 km^2 study area). The estimation of the spatial variability of recharge requires a higher level of hydraulic parameter information (e.g., >200 K values needed for an average absolute recharge error <20%). The use of both SGD (as a calibration target) and pumping data are shown to be of benefit to the estimation of average recharge, particularly where relatively few preferred hydraulic parameter values are included. The addition of pumping data also produces improved estimates of the spatial variability in recharge.

This study offers guidance to groundwater modelling practitioners by highlighting the potential effects of non-uniqueness in terms of recharge estimability in a real-world modelling scenario, and by providing insight into scenarios under which recharge may (or may not) be informed through model calibration in practical situations. The findings of this study provide a useful benchmark for evaluating the extent to which field-scale groundwater models can be used to inform recharge under practical data-availability limitations. Future work in this area should evaluate the estimability of transient recharge through calibration, and consider field sites that display different characteristics (e.g., surface water-groundwater interactions).

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